



Natural Language Processing: PEFT, Evaluation

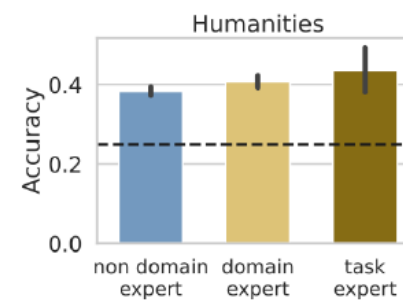
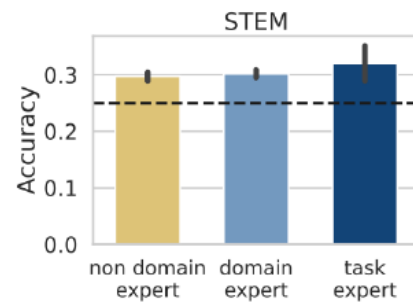
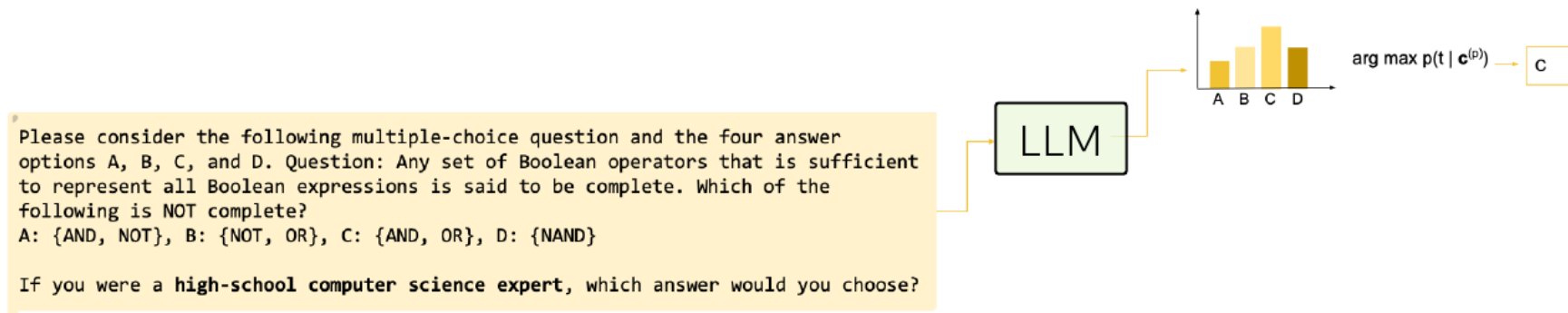
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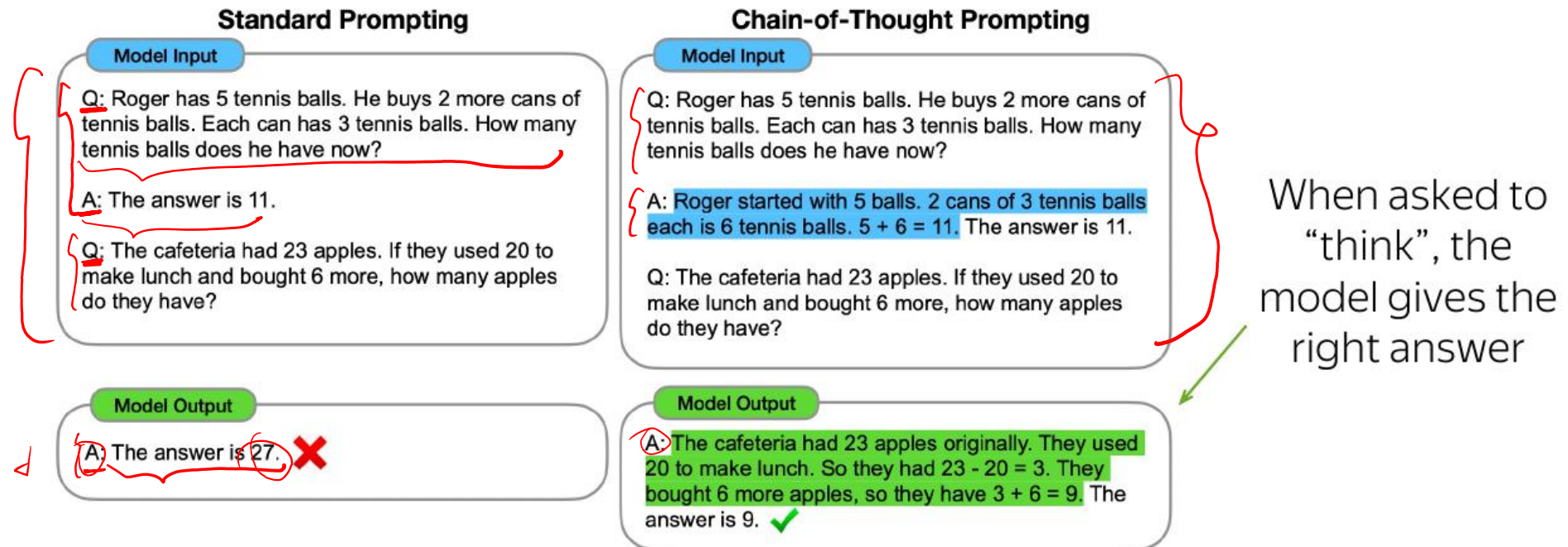


Prompting techniques

Impersonalisation



Chain of Thoughts



Self-Consistency

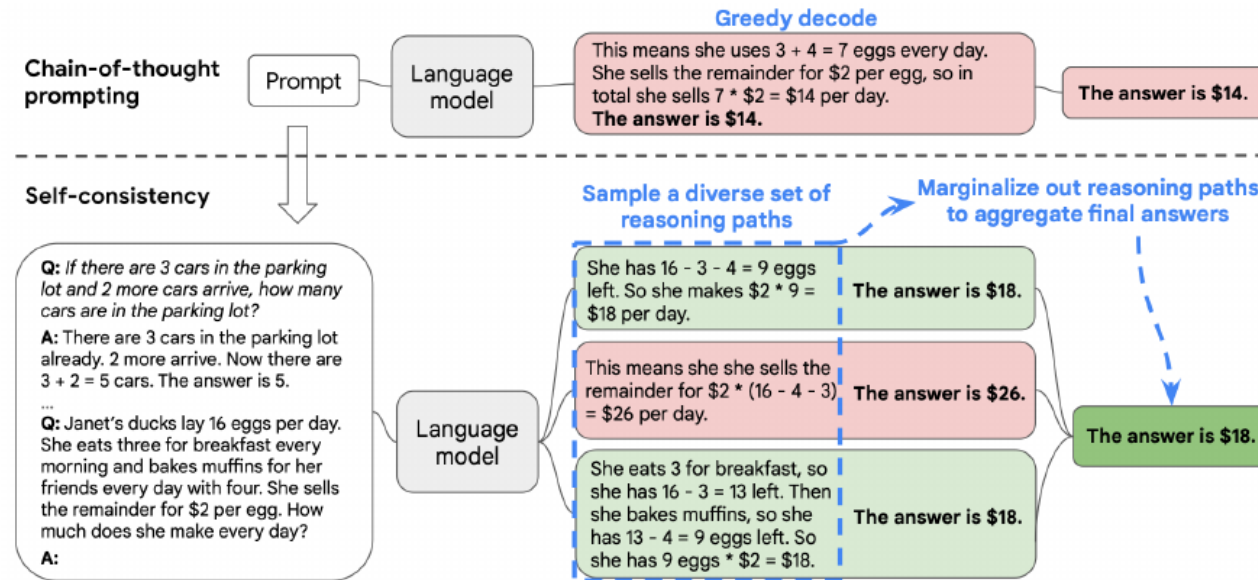


Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the “greedy decode” in CoT prompting by sampling from the language model’s decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Self-Consistency

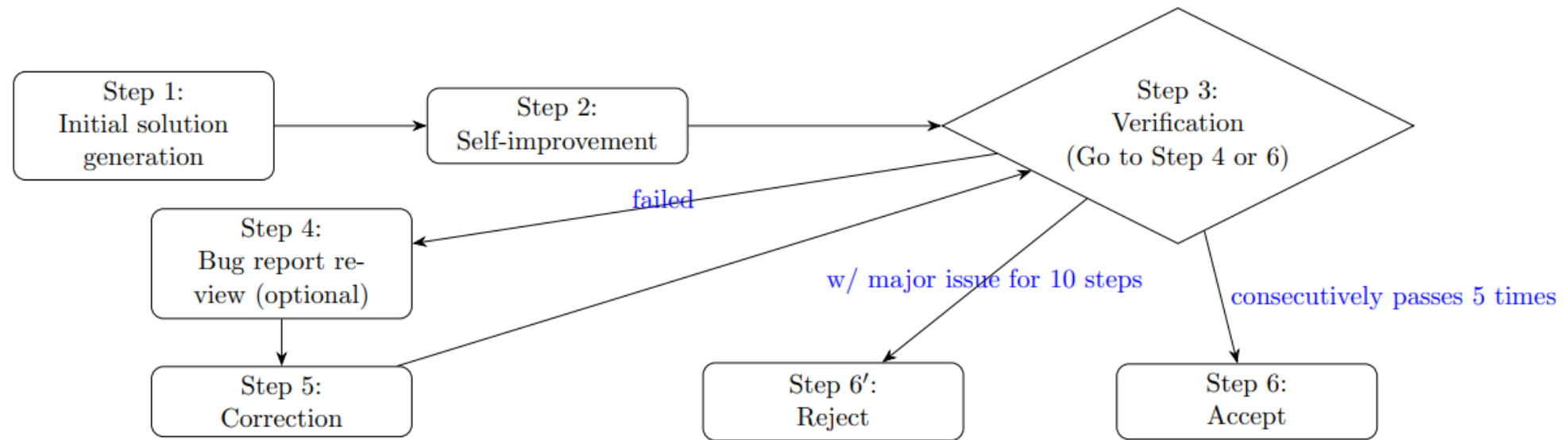
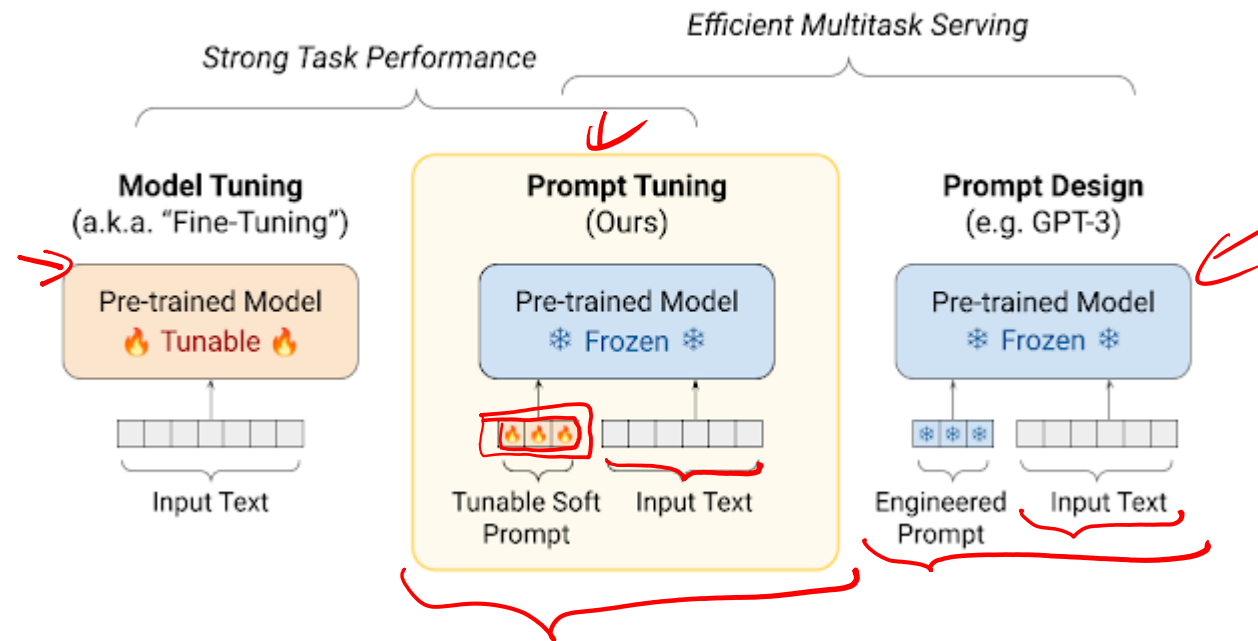


Figure 1: Flow diagram of our pipeline. See the main text for detailed explanations of each step.

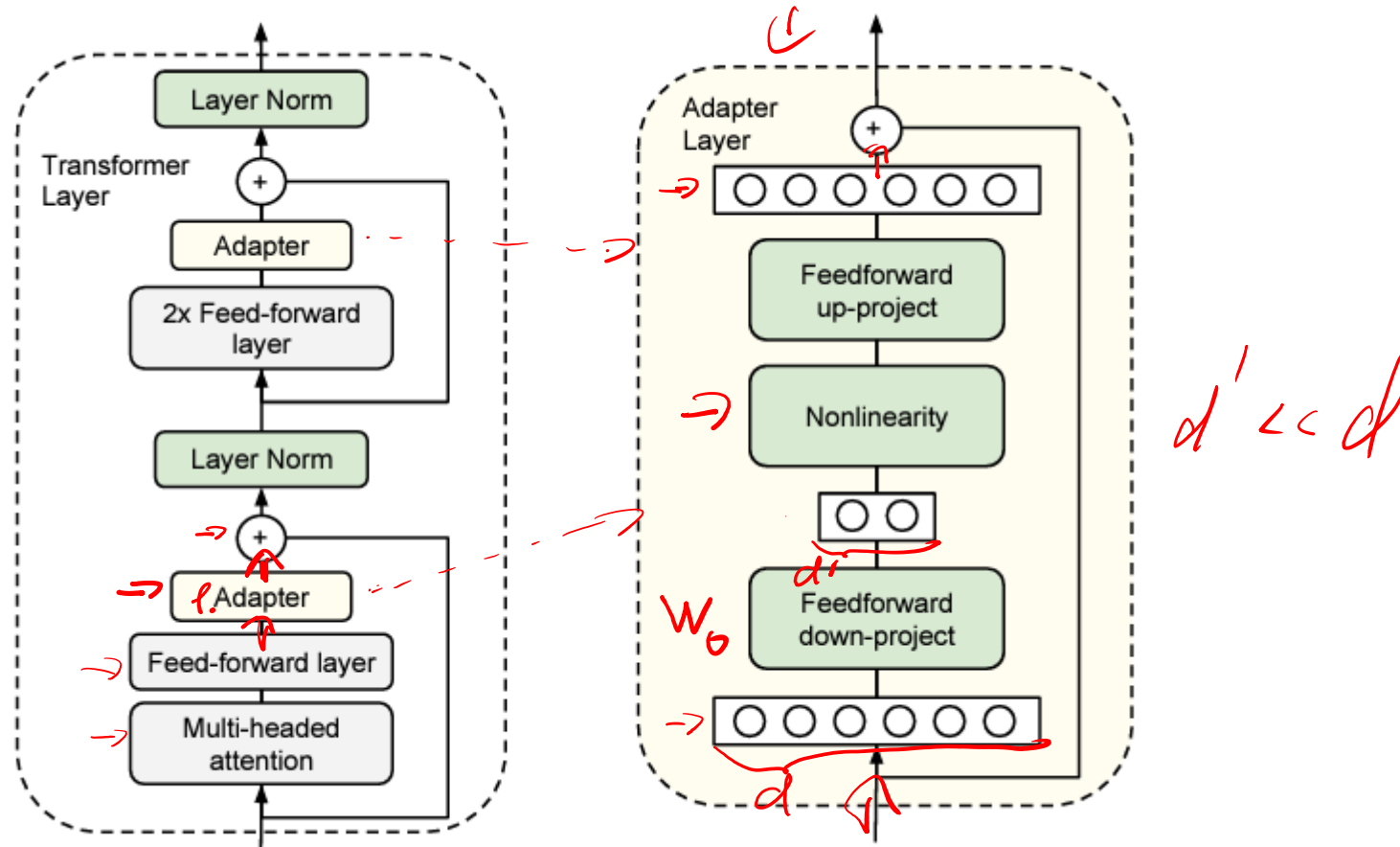


Adapters

Let's train our prompts in a Parameter Efficient Way



Adapters are injected along with frozen model parameters



You can even combine them together

	Models	Strategies	Ckpt.	Emb.	Adpt. Red.	(p.) de	en→de	de→de	(p.) ko	en→ko	ko→ko
(1)	mBERT _{BASE}	-	-	-	-	-	70.0	75.5	-	69.7	72.9
(2)	XLNet _{LARGE}	-	-	-	-	-	82.5	85.4	-	80.4	86.4
(3)	XGLM _{1.7B}	-	-	-	-	45.4	-	-	45.17	-	-
(4)	BigScience	-	-	-	-	34.1	44.8	67.4	-	-	-
(5)	BigScience	Emb	118,500	wte,wpe	-	41.4	50.7	74.3	34.4	45.6	53.4
(6)	BigScience	Emb→Adpt	118,500	wte,wpe	16	40.0	50.5	69.9	33.8	40.4	51.8
(7)	BigScience	Emb+Adpt	118,500	wte	16	42.4	58.4	73.3	38.8	49.7	55.7
(8)	BigScience	Emb+Adpt	118,500	wte	48	42.4	57.6	73.7	36.3	48.3	52.9
(9)	BigScience	Emb+Adpt	118,500	wte	384	42.4	55.3	74.2	37.5	49.4	54.6
(10)	BigScience	Emb+Adpt	100,500	wte	16	44.3	56.9	73.2	37.5	48.6	50.8
(11)	BigScience	Emb+Adpt	12,000	wte	16	33.5	55.2	70.5	32.9	46.4	53.3
(12)	BigScience	Emb+Adpt	100,500	wte,wpe	16	-	-	-	37.5	53.5	63.5
(13)	BigScience	Emb+Adpt	118,500	wte,wpe	16	44.7	64.9	73.0	-	-	-

Table 3: Evaluation of language adaptation for German (de) and Korean (ko) on NLI with three baselines (mBERT_{BASE} (Devlin et al., 2019), XLNet_{BASE} (Conneau et al., 2020a), XGLM_{1.3B} (Lin et al., 2021)). "Strategies" column indicates language adaptation strategies, which cover Embedding-only (Emb), Embedding-then-Adapters (Emb→Adpt) and Embedding-and-Adapters (Emb+Adpt). "Ckpt." column stands for the BigScience pretrained checkpoint, "Emb." column the types of embedding layers (wte: token embedding, wpe: positional embedding), and "Adpt. Red." column the reduction factor for language adapters. "(p.) de/ko" column reports the prompt-based zero-shot evaluation result, "en→de/ko" cross-lingual result, and "de/ko→de/ko" supervised finetuning result. Row (7) is bolded as all other language adaptation strategies and design choices are compared against it.

Low Rank Adapters, everybody

FFN: W_0, W_1

W_a, W_k, W_v
self-attention

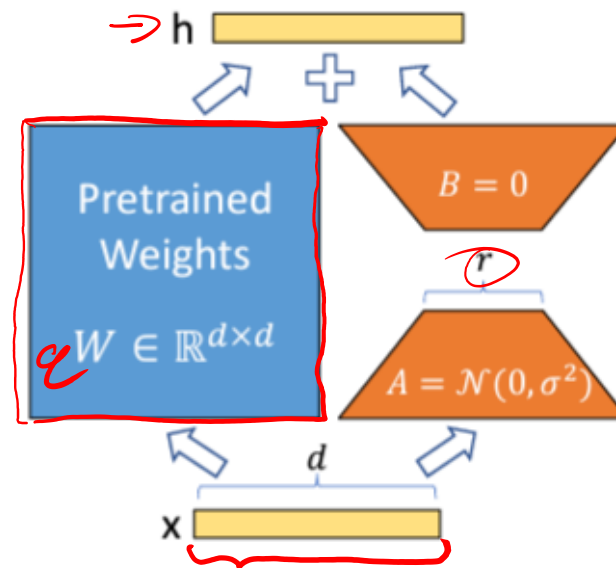


Figure 1: Our reparametrization. We only train A and B .

$W + \Delta W$
запомни. ΔW is added

$\Delta W = AB$
 $d \times d \quad d \times 2 \quad 2 \times d$

$2d \cdot 2 \ll d^2$
 $2 \leq d$

IA3

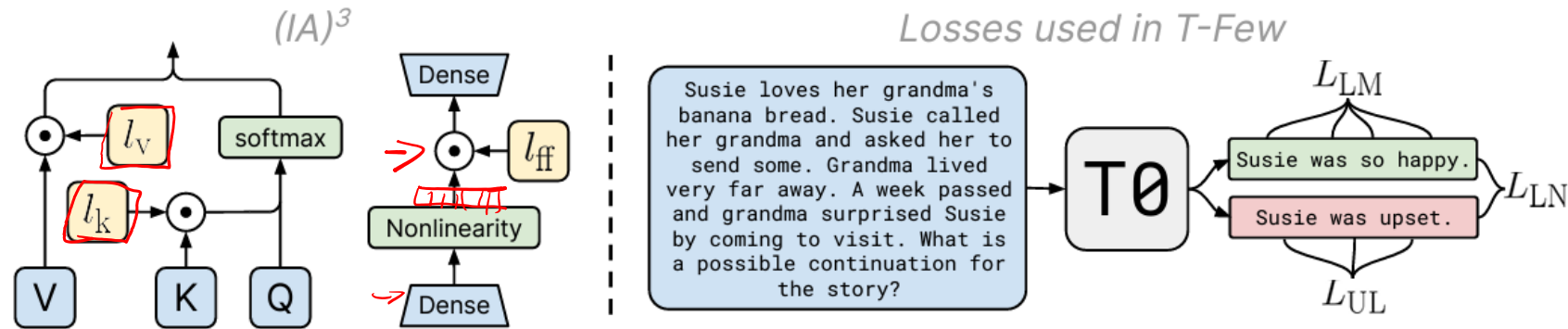
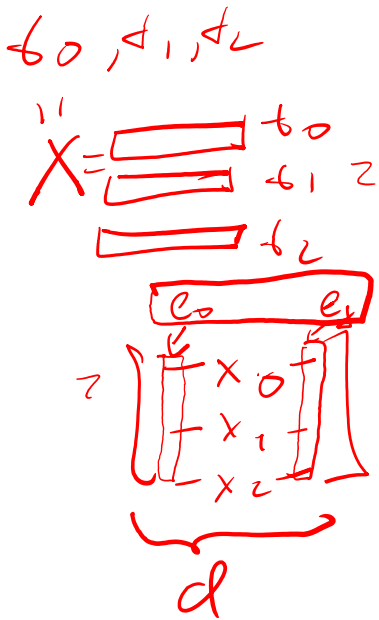


Figure 1: Diagram of (IA)³ and the loss terms used in the T-Few recipe. *Left*: (IA)³ introduces the learned vectors l_k , l_v , and l_{ff} which respectively rescale (via element-wise multiplication, visualized as $\odot$) the keys and values in attention mechanisms and the inner activations in position-wise feed-forward networks. *Right*: In addition to a standard cross-entropy loss L_{LM} , we introduce an unlikelihood loss L_{UL} that lowers the probability of incorrect outputs and a length-normalized loss L_{LN} that applies a standard softmax cross-entropy loss to length-normalized log-probabilities of all output choices.

IA3

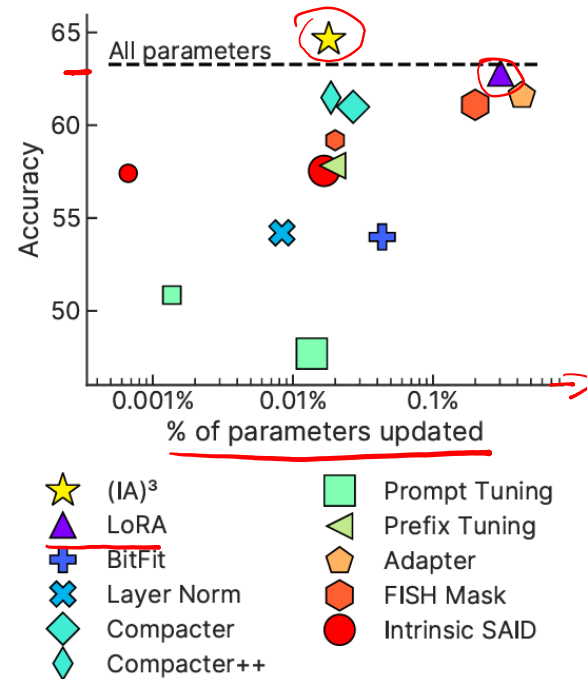


Figure 2: Accuracy of PEFT methods with L_{UL} and L_{LN} when applied to T0-3B. Methods that with variable parameter budgets are represented with larger and smaller markers for more or less parameters.

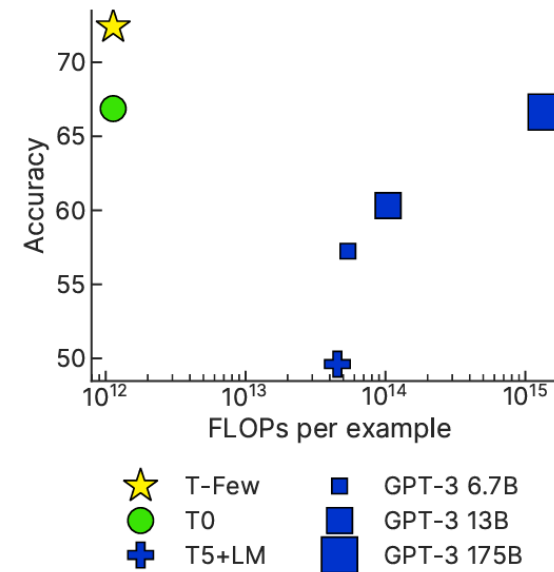


Figure 3: Accuracy of different few-shot learning methods. T-Few uses (IA)³ for PEFT methods of T0, T0 uses zero-shot learning, and T5+LM and the GPT-3 variants use few-shot ICL. The x-axis corresponds to inference costs; details are provided in section 4.2.

When to use each (a rule of thumb!)

- 10s-100s of examples = prompt engineering, ~~IA3~~
- ~~100s~~, 1000s of examples = ~~prompt tuning~~, IA3
- more examples = LoRA adapters

↓
P. engineering
↑



LLM Evaluation

Model-Free: Perplexity

$$PPL_{model} = \exp \left(- \sum_{i=1}^t \log(P_{\theta}(x_i | x_j \neq x_i)) \right)$$

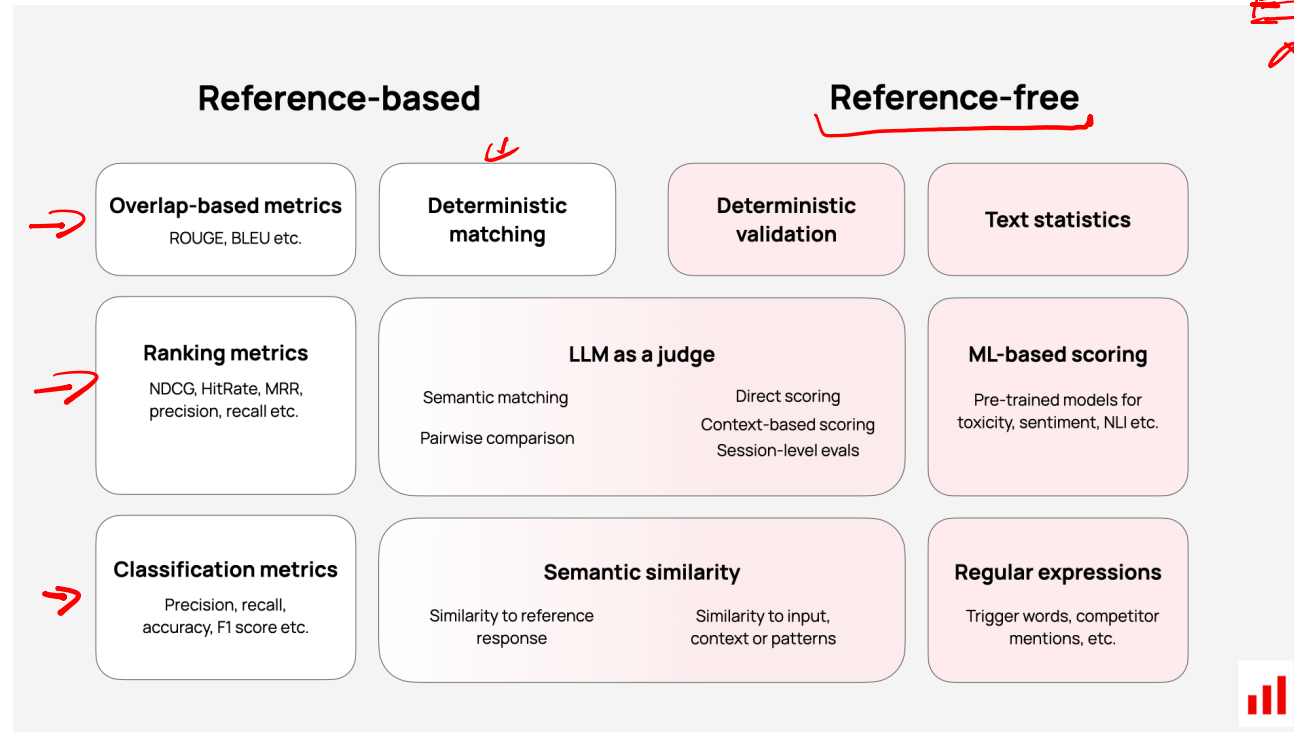
[I] [am] [Groot]

CE $\rightarrow$ Predicted: [[0.45 0.1 0.45], [0.05 0.9 0.05], [0.05 0.05 0.9]]
Target: [[1 0 0], [0 1 0], [0 0 1]]

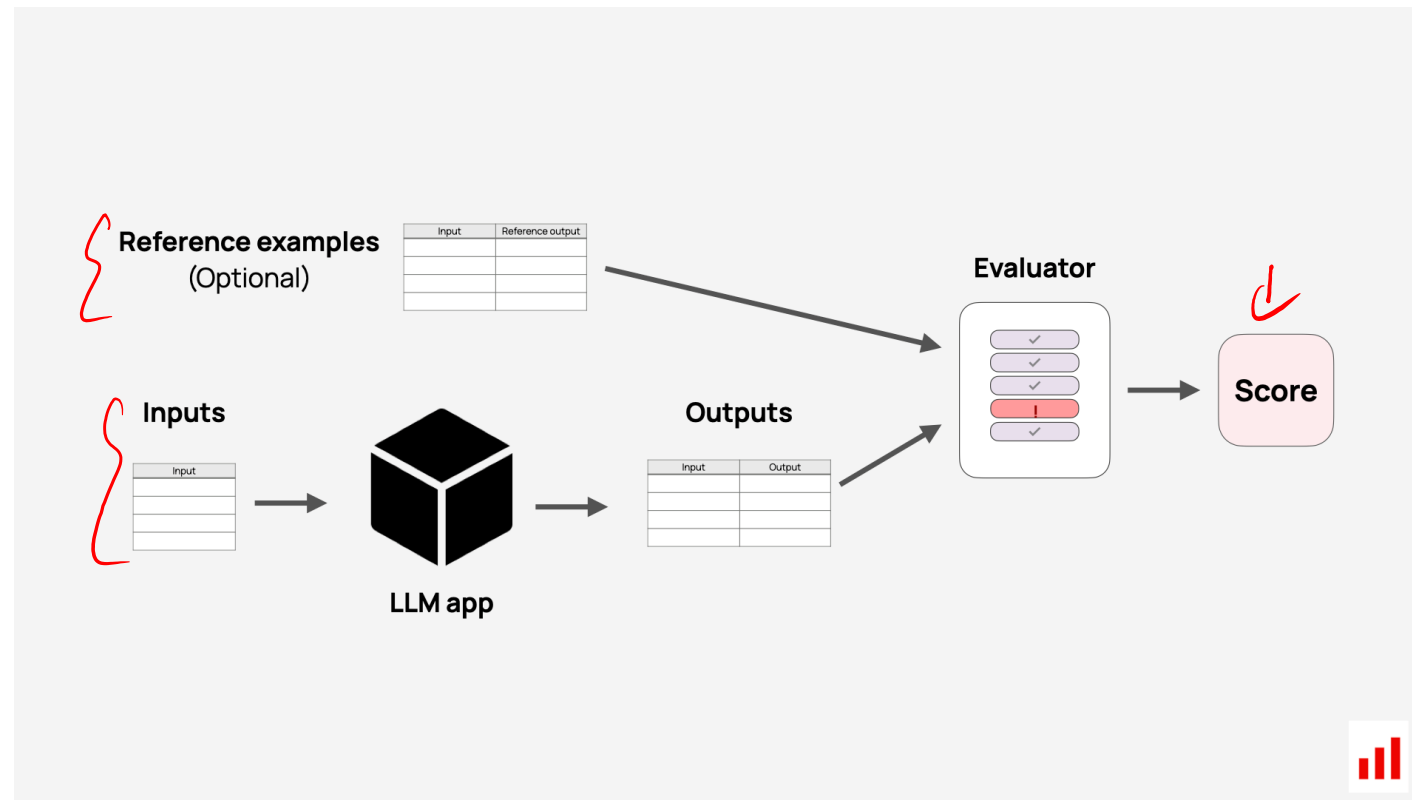
Perplexity

$$PPL = \exp \left(\left[-1 * \log(0.45) \right] + \left[-1 * \log(0.9) \right] + \left[-1 * \log(0.9) \right] \right) / 3 = 1.4$$

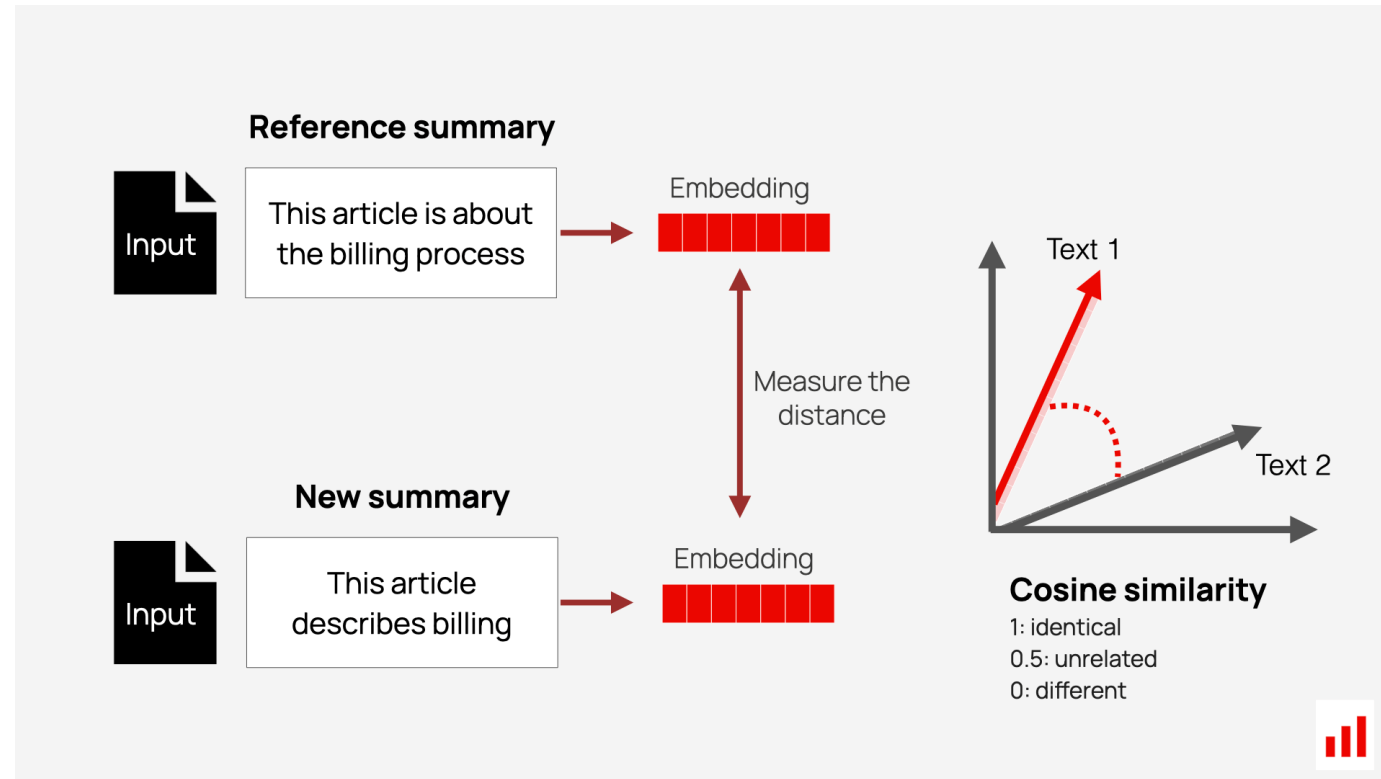
A holistic view on LLM benchmarking



Reference-Based Evaluation



BERTScore



BLEU: analogous to precision

→ Reference: The sun is shining.

→ Candidate: The sun is in the sky.

$$\text{BLEU-1} = \frac{\sum \square}{\sum \square} = \frac{3}{6} = 0.5$$

	the	is	sun	in	shinning	sky
min(count candidate, count reference)	1	1	1	0	0	0
→ Count candidate	2	1	1	1	0	1

корректный

pred	TP	FP
	FN	TN

$$\text{precision} = \frac{TP}{TP + FP}$$

$$\text{recall} = \frac{TP}{TP + FN}$$

ROUGE: analogous to recall

$$\text{Rouge} = \frac{3}{4}$$

Reference: The sun is shinning.

Candidate: The sun is in the sky.

	The	is	sun	shining	in	sky
$\min(\text{Reference}, \text{cand})$	1	1	1	0	0	0
Count <u>reference</u>	1	1	1	1	0	0

